POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2021
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a) Define Signal Analysis and Signal Processing. What are the advantages of Digital signal processing over Analog signal processing? 8

b) The output of an LTI system is y (n) = {2, 5, 1, -10, -10, -3, 6, 9}to an impulse response h(n) = {1, 2, 0, -3}.Find the input x(n) using convolution sum. The underline represents the element at n = 0. 7

a) Find the convolution between two signals 7
$x[n] = u[n]$ and $h[n] = a^n u[n],\ 0 < a < 1$

b) Determine Z–transform along with ROC of 8

$$x(n) = \left[\left(\frac{1}{2}\right)^n + \left(\frac{3}{4}\right)^n\right] u(n-5)$$

3. a) Find the 8-point DITFFT of $x(n) = sin\frac{3\pi n}{4}$, $for\ 0 \le n \le 7$ 8

b) What is Zero Padding ? Find the circular Convolution of $x_1[n] = \{1,2\}$ & $x_2[n] = \{3,2,1\}$ 7

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4. a) Obtain the parallel form realization of following IIR filter. 7

$$H(z) = \frac{3(2z^2 + 5z + 4)}{(2z + 1)(z + 2)}$$

b) Compute Lattice coefficients and draw lattice structure for given IIR system. Also check stability of the system. 8
$H(z) = 1 / (1 - 0.525z^{-1} + 0.6125z^{-2} + 0.3z^{-3})$

5. a) Design a lowpass filter which will have -3 dB cut off at 30π rad/sec and an attenuation of 50 dB at 45π rad/sec. The filter is required to have a linear phase and the system uses a sampling rate of 100 samples/sec. 8

b) Design a linear FIR filter using Kaiser window to meet the following specifications: 7

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$$0.99 \leq |H(e^{j\omega})| \leq 1.01, \quad for \ 0 \leq |\omega| \leq 0.19\pi$$
$$\leq |H(e^{j\omega})| \leq 0.01, \quad for \ 0.21\pi \leq |\omega| \leq \pi$$

6. a) Determine H(z) using impulse invariance method for the following analog filter's transfer function :

$$H_a(s) = \frac{1}{(s + 0.5) + (s^2 + 0.5s + 2)}$$

b) Design a digital lowpass Butterworth filter using Bilinear Transformation method to meet the following specifications:
Passband attenuation = 1 dB
Stopband attenuation = 16 dB
Passband edge = 120 Hz
Stopband edge = 170 Hz
Sampling frequency = 256 Hz

7. Write short notes on: (**Any two**)
   a) Region of Convergence
   b) Symmetric and Anti-symmetric filter
   c) Kaiser window

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